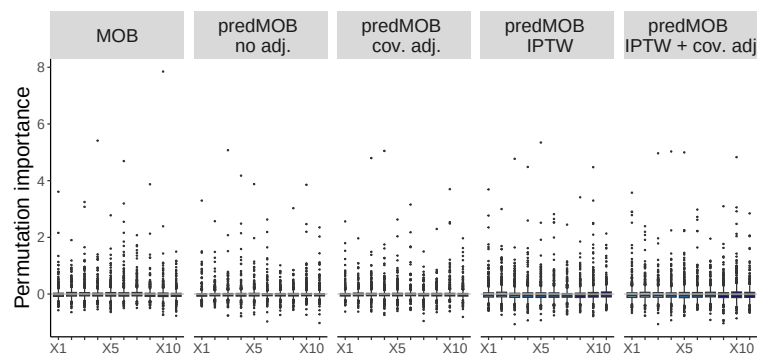


# Supplementary Material

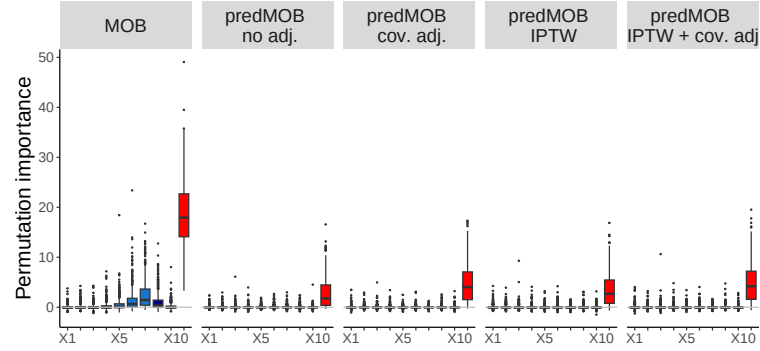
## S 1 Simulation results for the identification of predictive factors: additional scenarios from Table 1

We show the permutation importance results for MOB and predMOB in combination with various adjustment methods in simulation settings from Table 1 that were not presented in the main manuscript. As in Figure 2 instrumental variables are shown in light blue, true confounders in medium blue and factors only associated with outcome in dark blue. The boxplot for a true predictive factor is highlighted in red.

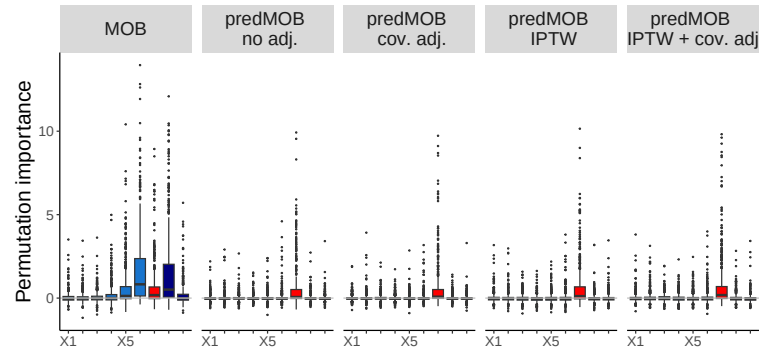
**Fig. S.1: Scenario 0:** Null scenario.



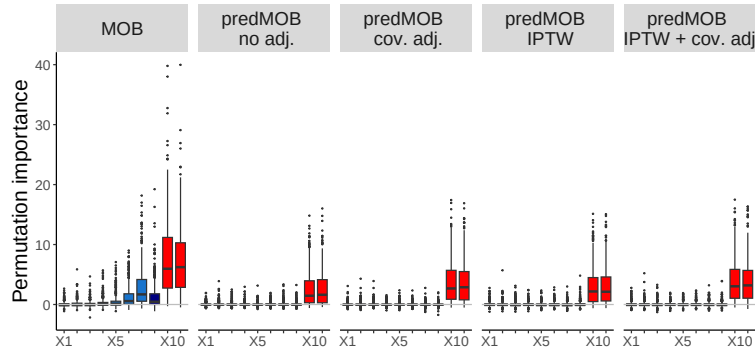
**Fig. S.2: Scenario B:**  $X_{10}$  has both a prognostic and a quantitative predictive effect



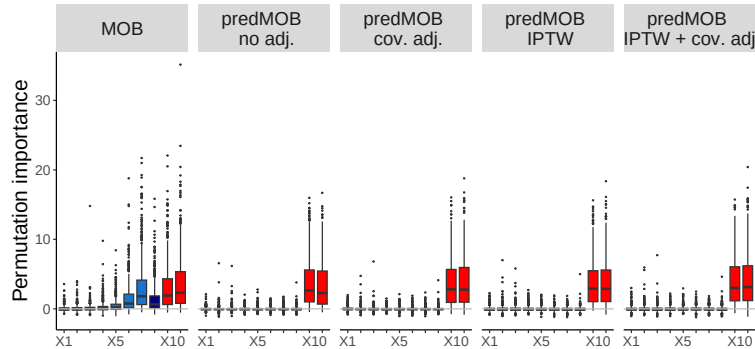
**Fig. S.3: Scenario G2:** Confounding variable  $X_7$  and predictive factor  $X_{10}$  are negatively correlated, however,  $X_{10}$  is not observed. Note:  $X_7$  is colored red even though it is not a predictive factor, but in this scenario it serves as a kind of surrogate, because it is correlated with an (unobserved) predictive factor.



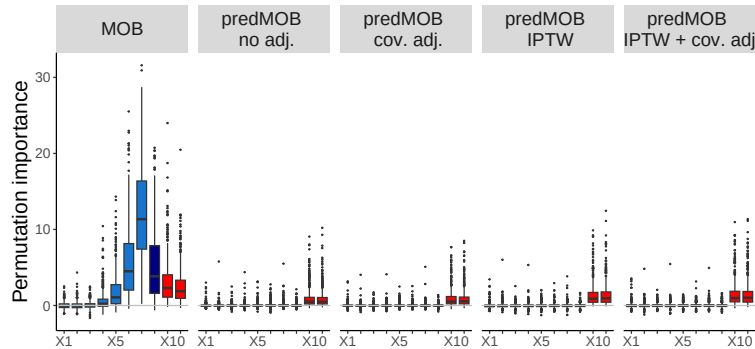
**Fig. S.4: Scenario H.1:**  $X_9$  and  $X_{10}$  both predictive only with predictive effect of  $X_9$  smaller than that of  $X_{10}$  (cf. Table 1).



**Fig. S.5: Scenario H.2:**  $X_9$  and  $X_{10}$  both prognostic and predictive with different effect sizes (cf. Table 1).



**Fig. S.6: Scenario I:** Higher order predictive pattern with three-way interaction of  $X_9$ ,  $X_{10}$  and treatment.



## S 2 Additional simulations depicting the need for covariate adjustment

In the simulations presented in the main part of the manuscript (Section 3) the true underlying model for treatment allocation was a linear combination of the observed variables and thus the propensity score model could easily fit the underlying relationships. In this section we present scenarios with a more complex structure such that the linear propensity score model does not adequately fit the underlying structure, thereby leading to false positive findings.

The simulation settings are as described in the simulation section of the manuscript. The binary treatment variable  $T \sim B(1, p)$  depends on biomarkers  $X_6$  and  $X_7$  via the following cases, which can also be expressed as a non-linear form:

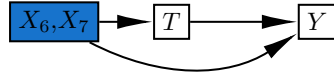
$$\mathbb{P}(T = 1) = \begin{cases} B(1, 0.5) & \text{if } X_7 = -1 \\ B(1, 0.1) & \text{if } X_7 = 1, X_6 = -1 \\ B(1, 0.9) & \text{if } X_7 = 1, X_6 = 1 \end{cases}$$

The binary distributed outcome variable  $Y \sim B(1, \mu)$  with expectation  $\mu$ :

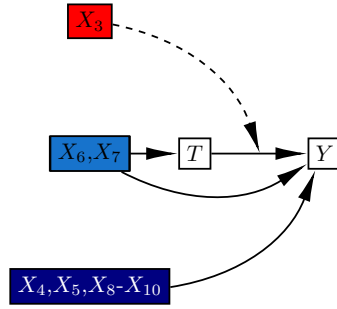
- **Scenario K:**  $\mu = 0.25 \cdot T + 0.8 \cdot X_6 + X_7$
- **Scenario L:**  $\mu = \text{base formula} + 0.5 \cdot X_3 \cdot T$
- **Scenario M:**  $\mu = \text{base formula} + 0.5 \cdot X_7 \cdot T$

with base formula =  $0.25 \cdot T + 0.1X_4 + 0.15X_5 + 0.2X_6 + 0.25X_7 + 0.2X_8 + 0.1X_9 + 0.15X_{10}$  as in the main simulations (Section 3).

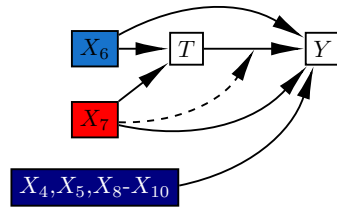
(a) Scenario K



(b) Scenario L

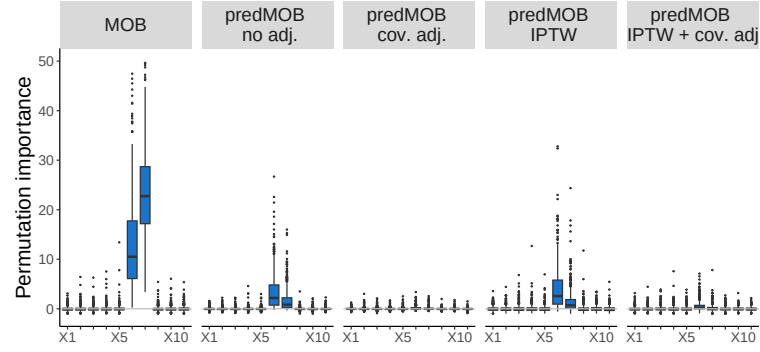


(c) Scenario M

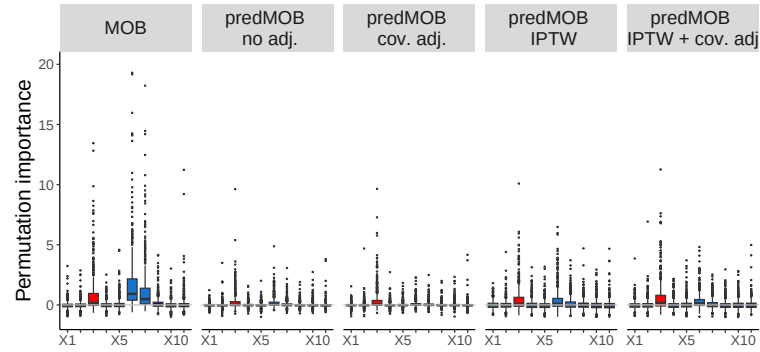


**Fig. S.7:** Graphical illustration of the additional simulation scenarios K-M. As in the main simulations effect-modification is marked as dashed arrows on the edge representing the treatment effect. Instrumental variables are shown in light blue, true confounders in medium blue and factors only associated with outcome in dark blue; predictive factors are highlighted in red.

**Fig. S.8: Scenario K:** no predictive variables.

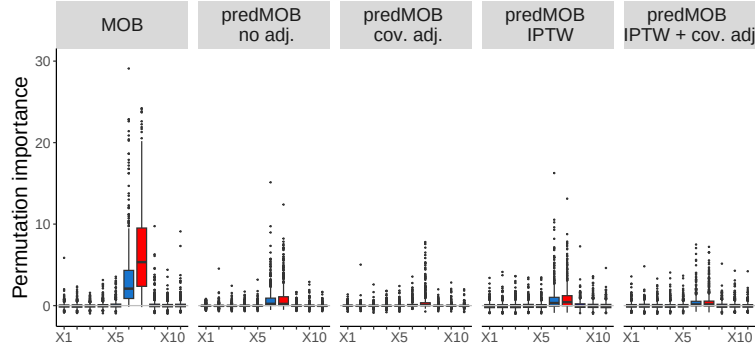


**Fig. S.9: Scenario L:** only  $X_3$  has a predictive effect.



Figures S.8 - S.10 show the results of these simulation scenarios in terms of the permutation importance (note: y-axis has been slightly cropped, thereby excluding a few outliers, for better visual presentation). Furthermore, rank-based presentations of these scenarios are depicted in Supplement Section S 6.1. We observe in Scenario K that without covariate adjustment variable  $X_6$  and  $X_7$  are incorrectly identified by predMOB as predictive factors based on the relatively high permutation importance

**Fig. S.10: Scenario M: only  $X_7$  has a predictive effect.**



values and consistent ranks 1 and 2. In Scenario L and M the respective predictive factors  $X_3$  and  $X_7$  are correctly identified in terms of relatively high permutation importance and often top rank, but also  $X_6$  shows an increased permutation importance and top rank even though it is not predictive. When using predMOB without covariate adjustment the influence of  $X_6$  and  $X_7$  on treatment assignment can lead to a false positive finding for these variables as predictive factors. When using IPTW we also have increased chances of false positives due to the fact that the true non-linear association in the treatment assignment is not well represented by the linear propensity score model that was fitted. In these scenarios predMOB with covariate adjustment shows the most promising results, followed by its combination with IPTW.

### S 3 Additional simulations depicting the need for IPTW

We show additional scenarios for cases where the true model underlying the treatment assignment is linear and the outcome model is not. In these scenarios the propensity score model can be used as a suitable approximation for the treatment assignment, but the linear model fitted for the outcome is not correct as it does not take into account the interaction.

Specifically we model the binary treatment variable  $T$  as in the main section of the simulations (Section 3) with  $T \sim B(1, p)$ ; depending on biomarkers  $X_1, \dots, X_7$  via the logistic regression model

$$\text{logit}(p) = \beta_0 + \log(1.1)X_1 - \log(1.2)X_2 + \log(1.3)X_3 - \log(1.1)X_4 + \log(1.2)X_5 - \log(1.3)X_6 + \log(1.4)X_7,$$

with  $\beta_0$  being chosen so  $p = 0.5$ ,

The binary distributed outcome variable  $Y \sim B(1, \pi)$  with  $\text{logit}(\pi) = \mu$ :

- **Scenario N:**

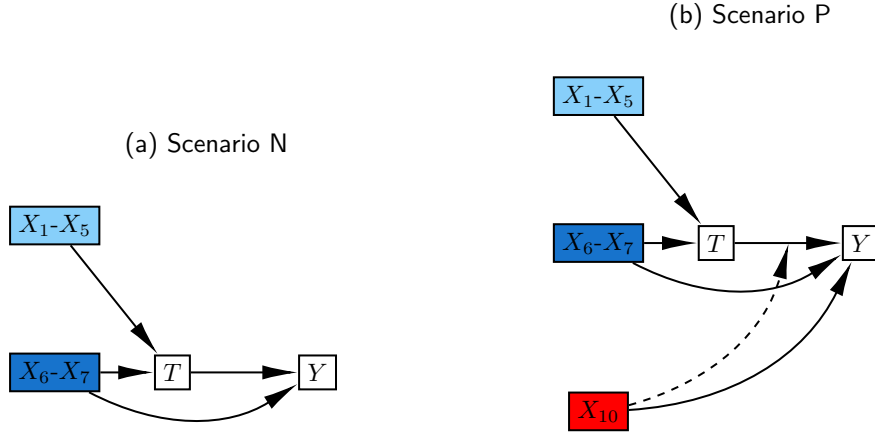
$$\mu_{\text{scenarioN}} = 0.5 \cdot T + \begin{cases} 0.75 & \text{if } X_7 = -1 \\ -0.8 & \text{if } X_7 = 1, X_6 = -1 \\ +1.5 & \text{if } X_7 = 1, X_6 = 1 \end{cases}$$

- **Scenario P:**  $\mu_{\text{scenarioP}} = \mu_{\text{scenarioN}} + 1.5 \cdot X_{10} \cdot T$

Figures S.12 - S.13 show the results of these simulation scenarios in terms of the permutation importance. See Appendix Section S 6.1 for rank-based results.

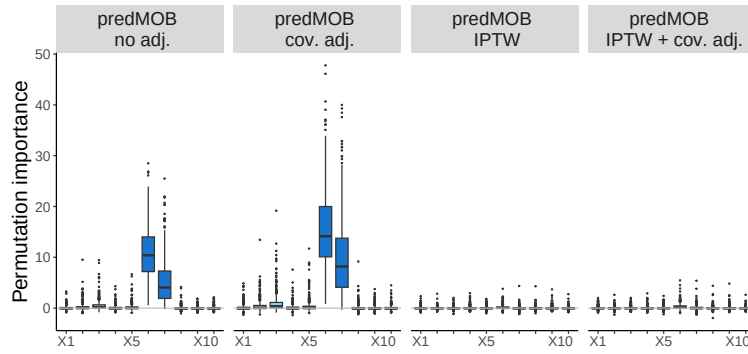
In scenarios N and P, where there is a non-linear relationship between variables  $X_6, X_7$  and the outcome, we observe that predMOB with no adjustment and covariate adjustment (which assumes a linear relationship) indicate false positive findings in terms of possible predictive effects for  $X_6, X_7$  represented by relatively high permutation importance values and consistently top ranks. The predMOB methods with IPTW have an advantage in these scenarios as the true propensity score model is in





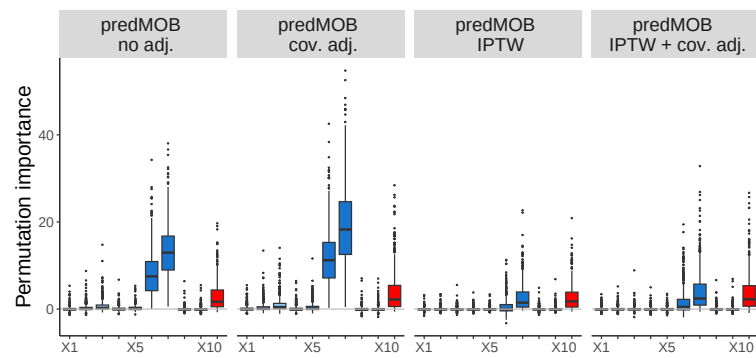
**Fig. S.11:** Graphical illustration of the additional simulation scenarios N and P. As in the main simulations effect-modification is marked as dashed arrows on the edge representing the treatment effect. Instrumental variables are shown in light blue, true confounders in medium blue and factors only associated with outcome in dark blue; predictive factors are highlighted in red.

**Fig. S.12: Scenario N:** no predictive variables.



fact linear and can thus be appropriately modeled with a linear fit leading to a reduced risk of false positives. In scenario P variable  $X_{10}$  has a predictive effect and can be correctly identified by all predMOB approaches in terms of relatively high permutation importance and top rank, albeit  $X_6$  and  $X_7$  are still falsely identified as potential variables with predictive effects.

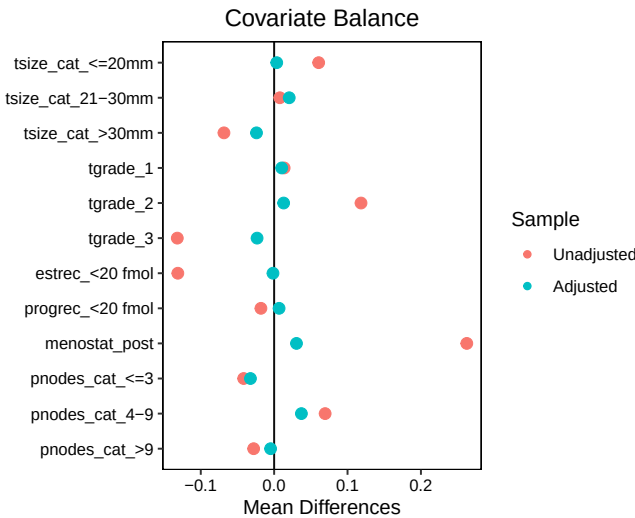
**Fig. S.13: Scenario P:** only  $X_{10}$  has a predictive effect.



# S 4 Additional outputs for application example of GBSG2 trial

Descriptive statistics for the two GBSG2 trial subpopulations by hormonal therapy yes/no are tabulated in Supplementary Table S.1.

Figure S.14 displays the absolute mean differences in the covariates between the two treatment arms in the non-randomized subpopulation. The red dots reveal that the covariates with the largest imbalance are menopausal status, estrogen receptor status and tumor grade. After using IPTW, the mean differences in the weighted population (marked in blue) are close to zero.



**Fig. S.14:** Comparison of difference in means between patients of the non-randomized population treated with or without tamoxifen for potential confounder variables. Red dots represent the unadjusted data while blue dots display the differences after IPTW. The differences in menopausal status, estrogen receptor status and tumor grade can be balanced out by IPTW.

**Table S.1:** Descriptive statistics for GBSG2 data by subpopulation and hormonal therapy

|                                    | Randomized        |                   |                   | Non-Randomized    |                   |                   |
|------------------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|                                    | no Tamoxifen      | Tamoxifen         | All               | no Tamoxifen      | Tamoxifen         | All               |
|                                    | (N=189)           | (N=184)           | (N=373)           | (N=176)           | (N=71)            | (N=247)           |
| <b>Age [years]</b>                 |                   |                   |                   |                   |                   |                   |
| Mean (SD)                          | 54.6 (9.76)       | 56.0 (9.34)       | 55.3 (9.57)       | 51.2 (9.85)       | 57.9 (9.56)       | 53.2 (10.2)       |
| Median [Min, Max]                  | 55.0 [25.0, 80.0] | 58.0 [33.0, 72.0] | 57.0 [25.0, 80.0] | 50.0 [21.0, 76.0] | 58.0 [32.0, 80.0] | 52.0 [21.0, 80.0] |
| <b>Chemotherapy</b>                |                   |                   |                   |                   |                   |                   |
| 3xCMF                              | 93 (49.2%)        | 93 (50.5%)        | 186 (49.9%)       | 72 (40.9%)        | 42 (59.2%)        | 114 (46.2%)       |
| 6xCMF                              | 96 (50.8%)        | 91 (49.5%)        | 187 (50.1%)       | 104 (59.1%)       | 29 (40.8%)        | 133 (53.8%)       |
| <b>Number positive lymph nodes</b> |                   |                   |                   |                   |                   |                   |
| ≤3                                 | 115 (60.8%)       | 95 (51.6%)        | 210 (56.3%)       | 92 (52.3%)        | 37 (52.1%)        | 129 (52.2%)       |
| 4-9                                | 56 (29.6%)        | 62 (33.7%)        | 118 (31.6%)       | 50 (28.4%)        | 23 (32.4%)        | 73 (29.6%)        |
| >9                                 | 16 (8.5%)         | 27 (14.7%)        | 43 (11.5%)        | 32 (18.2%)        | 10 (14.1%)        | 42 (17.0%)        |
| Missing                            | 2 (1.1%)          | 0 (0%)            | 2 (0.5%)          | 2 (1.1%)          | 1 (1.4%)          | 3 (1.2%)          |
| <b>Menopausal Status</b>           |                   |                   |                   |                   |                   |                   |
| pre                                | 55 (29.1%)        | 45 (24.5%)        | 100 (26.8%)       | 89 (50.6%)        | 17 (23.9%)        | 106 (42.9%)       |
| post                               | 134 (70.9%)       | 139 (75.5%)       | 273 (73.2%)       | 87 (49.4%)        | 54 (76.1%)        | 141 (57.1%)       |
| <b>Tumor Size [mm]</b>             |                   |                   |                   |                   |                   |                   |

|       |                                        |             |             |             |             |            |             |
|-------|----------------------------------------|-------------|-------------|-------------|-------------|------------|-------------|
| ε1ppp | ≤20 mm                                 | 55 (29.1%)  | 48 (26.1%)  | 103 (27.6%) | 39 (22.2%)  | 20 (28.2%) | 59 (23.9%)  |
|       | 21-30 mm                               | 74 (39.2%)  | 79 (42.9%)  | 153 (41.0%) | 76 (43.2%)  | 30 (42.3%) | 106 (42.9%) |
|       | >30 mm                                 | 59 (31.2%)  | 56 (30.4%)  | 115 (30.8%) | 61 (34.7%)  | 21 (29.6%) | 82 (33.2%)  |
|       | Missing                                | 1 (0.5%)    | 1 (0.5%)    | 2 (0.5%)    | 0 (0%)      | 0 (0%)     | 0 (0%)      |
|       | <b>Tumor grade</b>                     |             |             |             |             |            |             |
|       | 1                                      | 22 (11.6%)  | 25 (13.6%)  | 47 (12.6%)  | 21 (11.9%)  | 9 (12.7%)  | 30 (12.1%)  |
|       | 2                                      | 118 (62.4%) | 118 (64.1%) | 236 (63.3%) | 102 (58.0%) | 51 (71.8%) | 153 (61.9%) |
|       | 3                                      | 43 (22.8%)  | 41 (22.3%)  | 84 (22.5%)  | 51 (29.0%)  | 11 (15.5%) | 62 (25.1%)  |
|       | Missing                                | 6 (3.2%)    | 0 (0%)      | 6 (1.6%)    | 2 (1.1%)    | 0 (0%)     | 2 (0.8%)    |
|       | <b>Estrogen receptors [fmol/l]</b>     |             |             |             |             |            |             |
|       | ≥20 fmol                               | 116 (61.4%) | 109 (59.2%) | 225 (60.3%) | 104 (59.1%) | 53 (74.6%) | 157 (63.6%) |
|       | <20 fmol                               | 68 (36.0%)  | 71 (38.6%)  | 139 (37.3%) | 69 (39.2%)  | 17 (23.9%) | 86 (34.8%)  |
|       | Missing                                | 5 (2.6%)    | 4 (2.2%)    | 9 (2.4%)    | 3 (1.7%)    | 1 (1.4%)   | 4 (1.6%)    |
|       | <b>Progesterone receptors [fmol/l]</b> |             |             |             |             |            |             |
|       | ≥20 fmol                               | 107 (56.6%) | 102 (55.4%) | 209 (56.0%) | 106 (60.2%) | 47 (66.2%) | 153 (61.9%) |
|       | <20 fmol                               | 75 (39.7%)  | 79 (42.9%)  | 154 (41.3%) | 66 (37.5%)  | 23 (32.4%) | 89 (36.0%)  |
|       | Missing                                | 7 (3.7%)    | 3 (1.6%)    | 10 (2.7%)   | 4 (2.3%)    | 1 (1.4%)   | 5 (2.0%)    |
|       | <b>RFS</b>                             |             |             |             |             |            |             |
|       | ≥ 2 years                              | 137 (72.5%) | 130 (70.7%) | 267 (71.6%) | 119 (67.6%) | 57 (80.3%) | 176 (71.3%) |
|       | < 2 years                              | 48 (25.4%)  | 46 (25.0%)  | 94 (25.2%)  | 48 (27.3%)  | 13 (18.3%) | 61 (24.7%)  |

|          |          |          |           |          |          |           |
|----------|----------|----------|-----------|----------|----------|-----------|
| Censored | 4 (2.1%) | 8 (4.3%) | 12 (3.2%) | 9 (5.1%) | 1 (1.4%) | 10 (4.0%) |
|----------|----------|----------|-----------|----------|----------|-----------|

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| Variable                                                             | estimate | lower | upper | pValue |
|----------------------------------------------------------------------|----------|-------|-------|--------|
| Tamoxifen treatment                                                  | -0.85    | -2.47 | 0.76  | 0.3    |
| Tumor size (21-30 mm)                                                | 0.13     | -0.83 | 1.10  | 0.787  |
| Tumor size ( $\geq 30$ mm)                                           | 0.41     | -0.58 | 1.40  | 0.415  |
| Progesterone receptor status ( $\leq 20$ fmol)                       | 0.85     | -0.18 | 1.87  | 0.104  |
| Estrogen receptor status ( $\leq 20$ fmol)                           | -0.51    | -1.60 | 0.58  | 0.357  |
| Post menopausal status                                               | -0.69    | -1.52 | 0.13  | 0.1    |
| Number positive lymph nodes (4-9)                                    | 1.04     | 0.24  | 1.83  | 0.011  |
| Number positive lymph nodes ( $\geq 9$ )                             | 1.32     | 0.11  | 2.53  | 0.033  |
| Tumor grade 2                                                        | 2.19     | 0.13  | 4.24  | 0.037  |
| Tumor grade 3                                                        | 2.67     | 0.55  | 4.79  | 0.014  |
| Tamoxifen treatment * Tumor size (21-30 mm)                          | -0.03    | -1.50 | 1.44  | 0.966  |
| Tamoxifen treatment * Tumor size ( $\geq 30$ mm)                     | -0.20    | -1.72 | 1.32  | 0.798  |
| Tamoxifen treatment * Progesterone receptor status ( $\leq 20$ fmol) | 0.67     | -0.79 | 2.14  | 0.367  |
| Tamoxifen treatment * Estrogen receptor status ( $\leq 20$ fmol)     | 0.41     | -1.12 | 1.94  | 0.599  |
| Tamoxifen treatment * Post menopausal status                         | 0.39     | -0.77 | 1.55  | 0.511  |
| Tamoxifen treatment * Number positive lymph nodes (4-9)              | -0.08    | -1.25 | 1.09  | 0.896  |
| Tamoxifen treatment * Number positive lymph nodes ( $\geq 9$ )       | 0.02     | -1.62 | 1.66  | 0.982  |

**Table S.2:** Results for randomized cohort applying logistic regression with all possible treatment interactions. Note: treatment interaction with tumor grade was excluded due to poor model fit. Abbreviations: lower/upper 95% confidence limits.

| Variable                                                             | estimate | lower | upper | pValue |
|----------------------------------------------------------------------|----------|-------|-------|--------|
| Tamoxifen treatment                                                  | -0.01    | -3.28 | 3.26  | 0.995  |
| Tumor size (21-30 mm)                                                | 1.43     | 0.30  | 2.55  | 0.013  |
| Tumor size ( $\geq 30$ mm)                                           | 1.24     | 0.07  | 2.42  | 0.039  |
| Tumor grade 2                                                        | -0.19    | -1.37 | 1.00  | 0.756  |
| Tumor grade 3                                                        | 0.10     | -1.41 | 1.61  | 0.898  |
| Progesterone receptor status ( $\leq 20$ fmol)                       | 0.63     | -0.23 | 1.50  | 0.149  |
| Estrogen receptor status ( $\leq 20$ fmol)                           | 0.61     | -0.26 | 1.49  | 0.166  |
| Post menopausal status                                               | 0.47     | -0.40 | 1.33  | 0.287  |
| Number positive lymph nodes (4-9)                                    | 0.07     | -0.81 | 0.95  | 0.877  |
| Number positive lymph nodes ( $\geq 9$ )                             | 1.53     | 0.50  | 2.55  | 0.004  |
| Tamoxifen treatment * Tumor size (21-30 mm)                          | -1.43    | -4.05 | 1.19  | 0.284  |
| Tamoxifen treatment * Tumor size ( $\geq 30$ mm)                     | -0.78    | -3.82 | 2.26  | 0.614  |
| Tamoxifen treatment * Tumor grade 2                                  | -1.05    | -4.30 | 2.19  | 0.523  |
| Tamoxifen treatment * Tumor grade 3                                  | 0.13     | -2.75 | 3.02  | 0.928  |
| Tamoxifen treatment * Progesterone receptor status ( $\leq 20$ fmol) | 2.83     | 0.23  | 5.43  | 0.033  |
| Tamoxifen treatment * Estrogen receptor status ( $\leq 20$ fmol)     | -1.29    | -3.51 | 0.93  | 0.254  |
| Tamoxifen treatment * Post menopausal status                         | -0.21    | -2.25 | 1.84  | 0.843  |
| Tamoxifen treatment * Number positive lymph nodes (4-9)              | 1.21     | -0.87 | 3.29  | 0.253  |
| Tamoxifen treatment * Number positive lymph nodes ( $\geq 9$ )       | -1.39    | -3.90 | 1.12  | 0.276  |

**Table S.3:** Results for non-randomized cohort applying logistic regression with all possible treatment interactions. Abbreviations: lower/upper 95% confidence limits.



## S 5 Additional outputs for application example of AMLSG 16-10 trial

| Variable                                 | estimate | lower | upper | pValue |
|------------------------------------------|----------|-------|-------|--------|
| AMLSG 16-10 (active treatment)           | -0.80    | -2.63 | 1.02  | 0.389  |
| Age                                      | -0.02    | -0.04 | -0.00 | 0.028  |
| Female sex                               | 0.37     | -0.07 | 0.80  | 0.096  |
| NPM1 mutated                             | 1.34     | 0.89  | 1.79  | <.001  |
| WBC (log10)                              | -0.57    | -0.98 | -0.16 | 0.007  |
| BM blasts                                | 0.11     | -1.04 | 1.26  | 0.847  |
| FLT3-ITD high                            | -0.46    | -1.02 | 0.11  | 0.113  |
| AMLSG 16-10 (active trt) * Age           | 0.01     | -0.02 | 0.04  | 0.426  |
| AMLSG 16-10 (active trt) * female sex    | -0.65    | -1.29 | -0.01 | 0.047  |
| AMLSG 16-10 (active trt) * NPM1 mutated  | 0.04     | -0.62 | 0.69  | 0.91   |
| AMLSG 16-10 (active trt) * WBC (log10)   | 0.38     | -0.20 | 0.96  | 0.196  |
| AMLSG 16-10 (active trt) * BM blasts     | 0.42     | -1.19 | 2.04  | 0.605  |
| AMLSG 16-10 (active trt) * FLT3-ITD high | 0.39     | -0.35 | 1.14  | 0.298  |

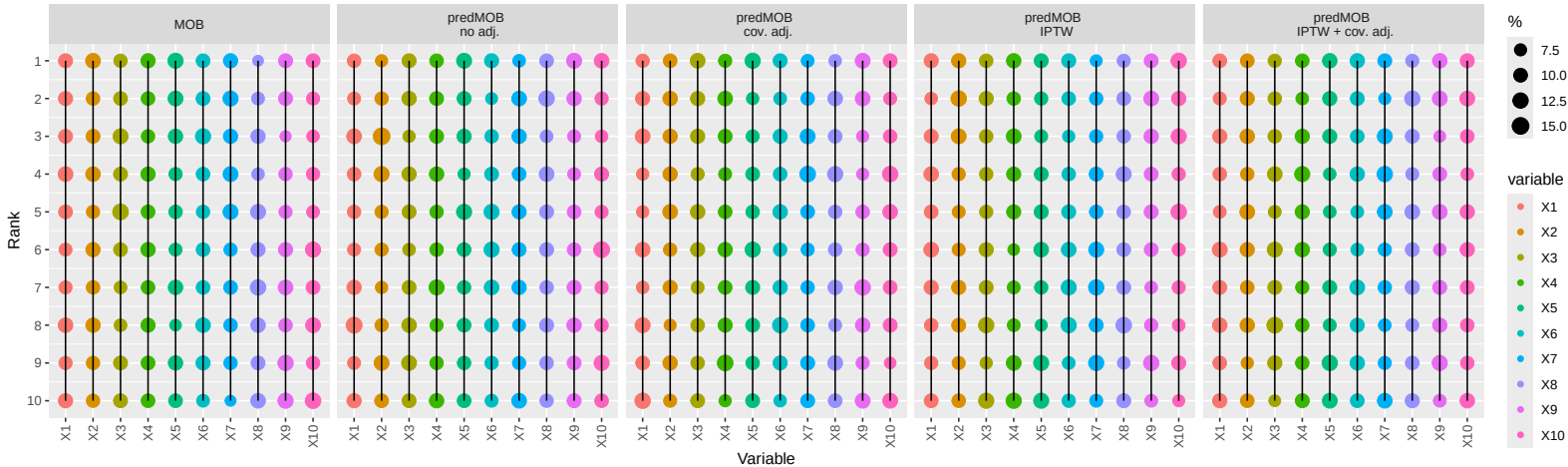
**Table S.4:** Results for response to induction therapy (CR/CRi) from logistic regression model with all treatment interactions. Abbreviations: BM, bone marrow; CR, complete remission; CRi, CR with incomplete hematologic recovery; ITD, internal tandem duplication; OR, odds ratio; WBC, white blood cells; lower/upper 95% confidence limits

## S 6 Rank-based presentation of results

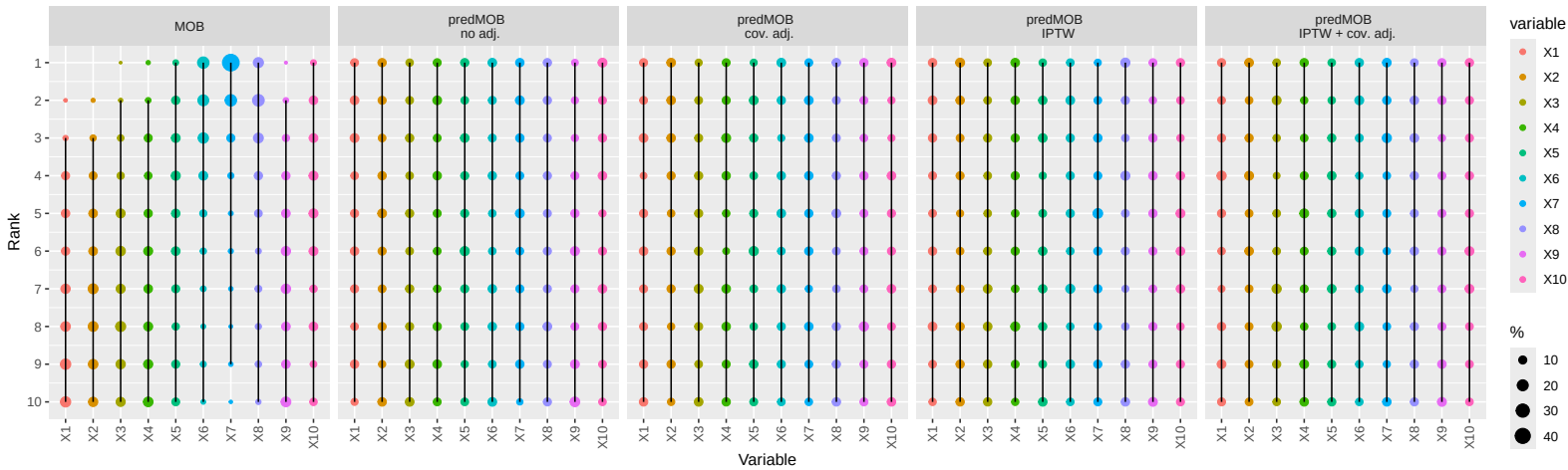
### S 6.1 Simulation results

In this section we present an alternative depiction of the simulation results where each variable is ranked by highest to smallest variable importance score across simulation runs for the different methods in various settings. The plots show the distribution of the ranks across the simulation runs (percentages of observed ranks presented by the size of dots as a measure of ranking stability) and the region between the 2.5% and the 97.5% percentile presented by the connecting line. Ties are given maximum rank. Note: ties occur almost exclusively for variable importance values equal to 0, meaning that the respective variable was not selected for any split in the tree, and thus has no prognostic/predictive effect.

Scenario 0  
 $\mu = 0$

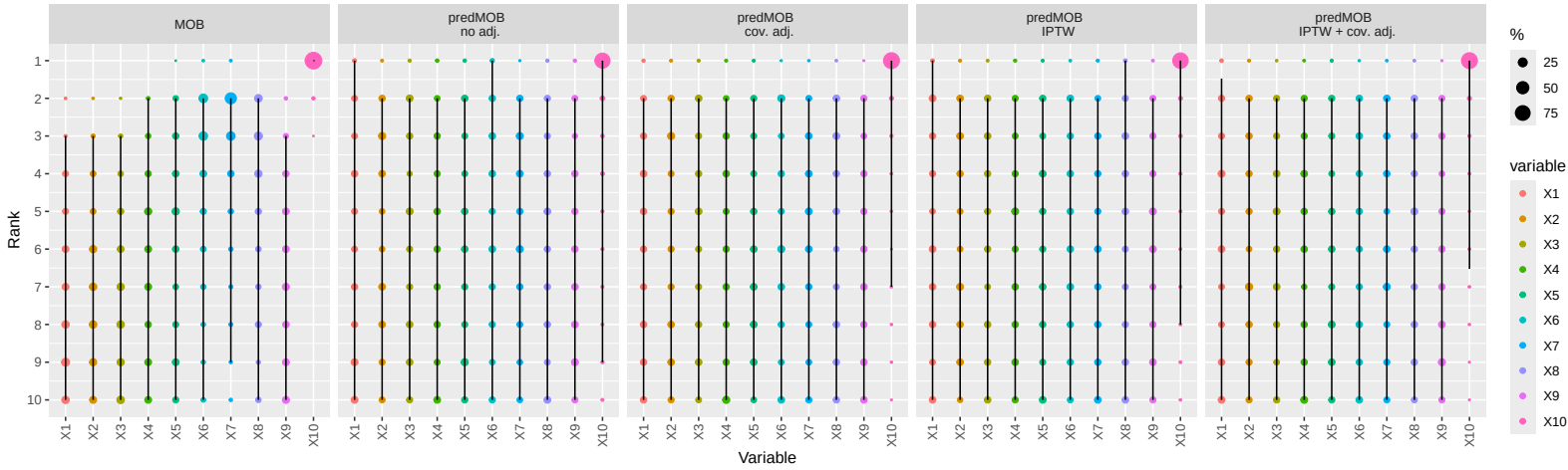


Scenario A1  
 $\mu = 0.25 \cdot \text{trt} + 0.1 \cdot X_4 + 0.15 \cdot X_5 + 0.2 \cdot X_6 + 0.25 \cdot X_7 + 0.2 \cdot X_8 + 0.1 \cdot X_9 + 0.15 \cdot X_{10}$

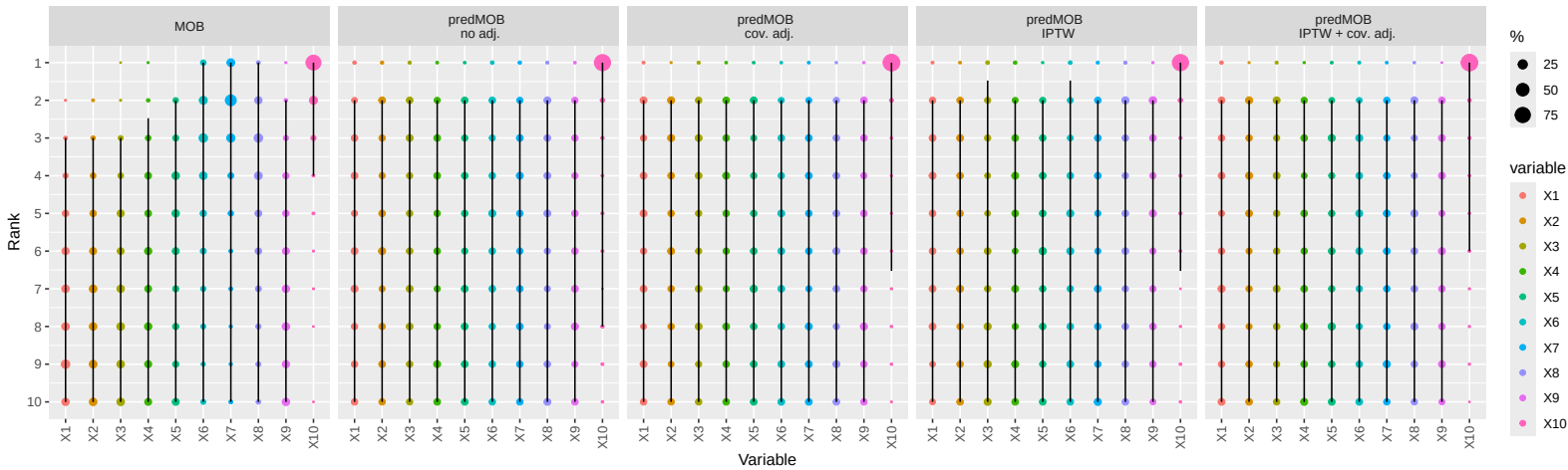


ddd20

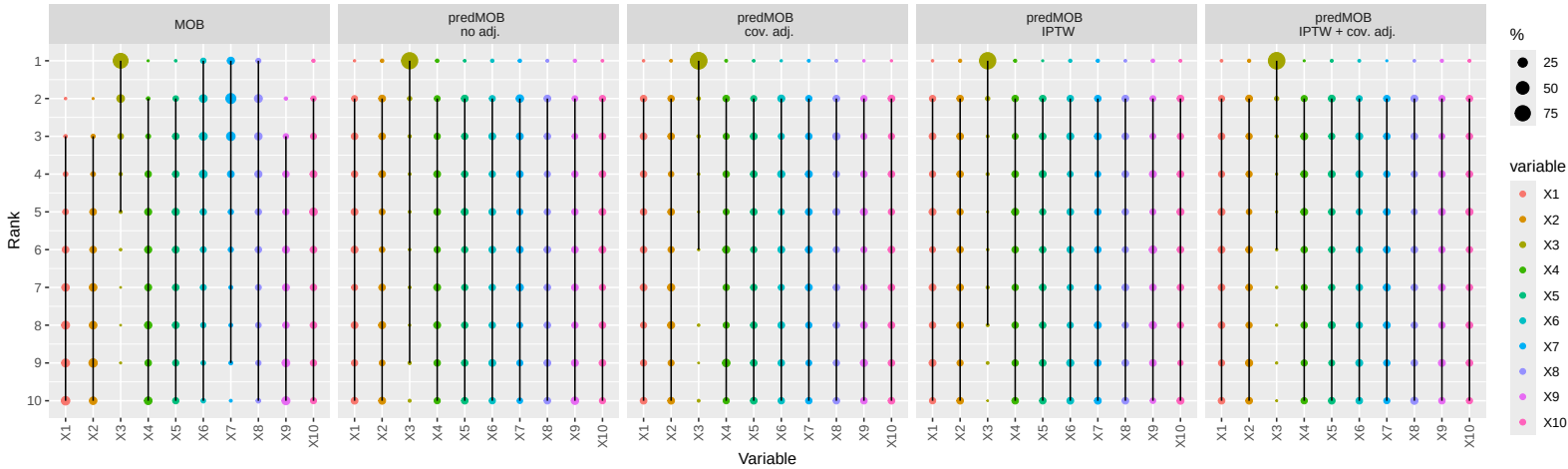
Scenario B  
 $\mu = 0.25 \cdot \text{trt} + 0.1 \cdot X_4 + 0.15 \cdot X_5 + 0.2 \cdot X_6 + 0.25 \cdot X_7 + 0.2 \cdot X_8 + 0.1 \cdot X_9 + 0.15 \cdot X_{10} + 0.5 \cdot X_{10} \cdot \text{trt}$



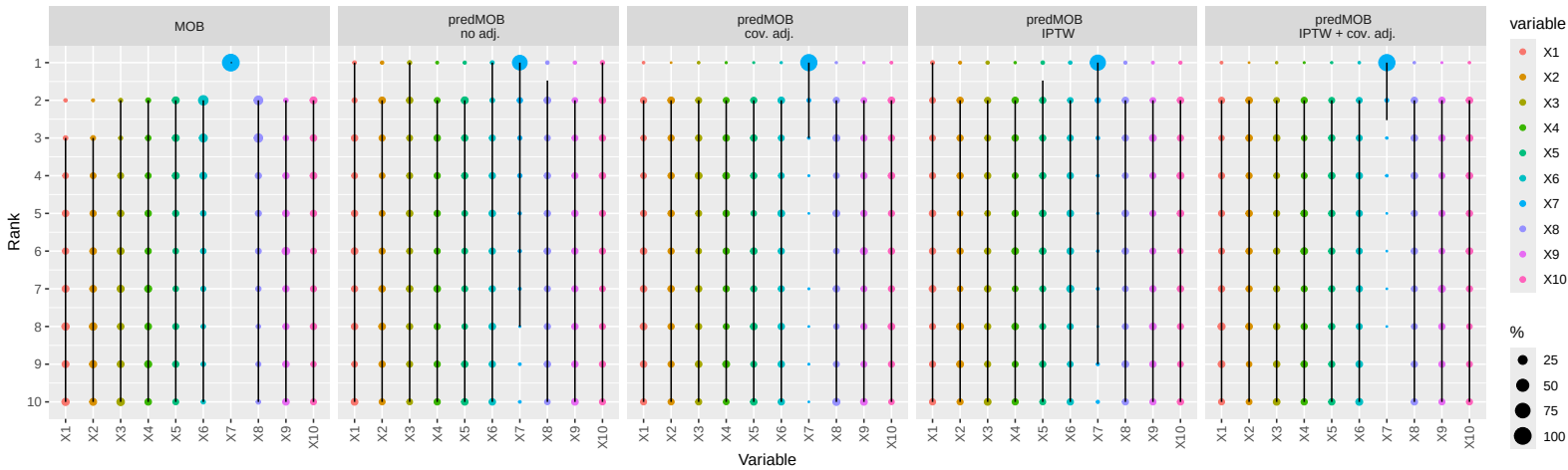
Scenario C  
 $\mu = 0.25 \cdot \text{trt} + 0.1 \cdot X_4 + 0.15 \cdot X_5 + 0.2 \cdot X_6 + 0.25 \cdot X_7 + 0.2 \cdot X_8 + 0.1 \cdot X_9 + 0.5 \cdot X_{10} \cdot \text{trt}$



Scenario D  
 $\mu = 0.25 \cdot \text{trt} + 0.1 \cdot X_4 + 0.15 \cdot X_5 + 0.2 \cdot X_6 + 0.25 \cdot X_7 + 0.2 \cdot X_8 + 0.1 \cdot X_9 + 0.15 \cdot X_{10} + 0.5 \cdot X_3 \cdot \text{trt}$



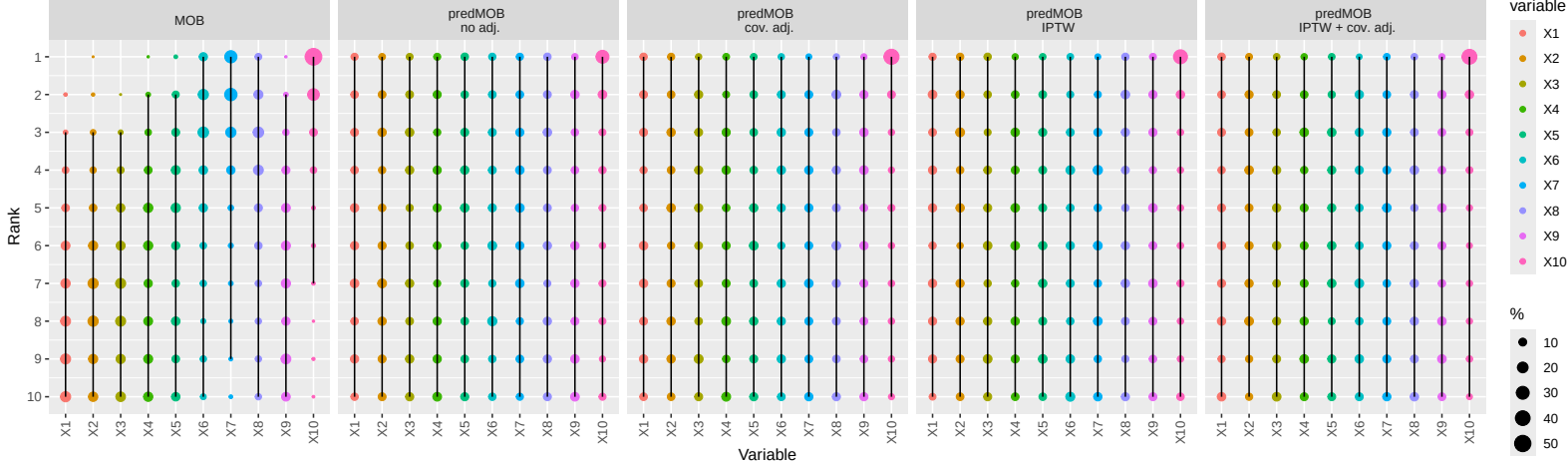
Scenario E  
 $\mu = 0.25 \cdot \text{trt} + 0.1 \cdot X_4 + 0.15 \cdot X_5 + 0.2 \cdot X_6 + 0.25 \cdot X_7 + 0.2 \cdot X_8 + 0.1 \cdot X_9 + 0.15 \cdot X_{10} + 0.5 \cdot X_7 \cdot \text{trt}$





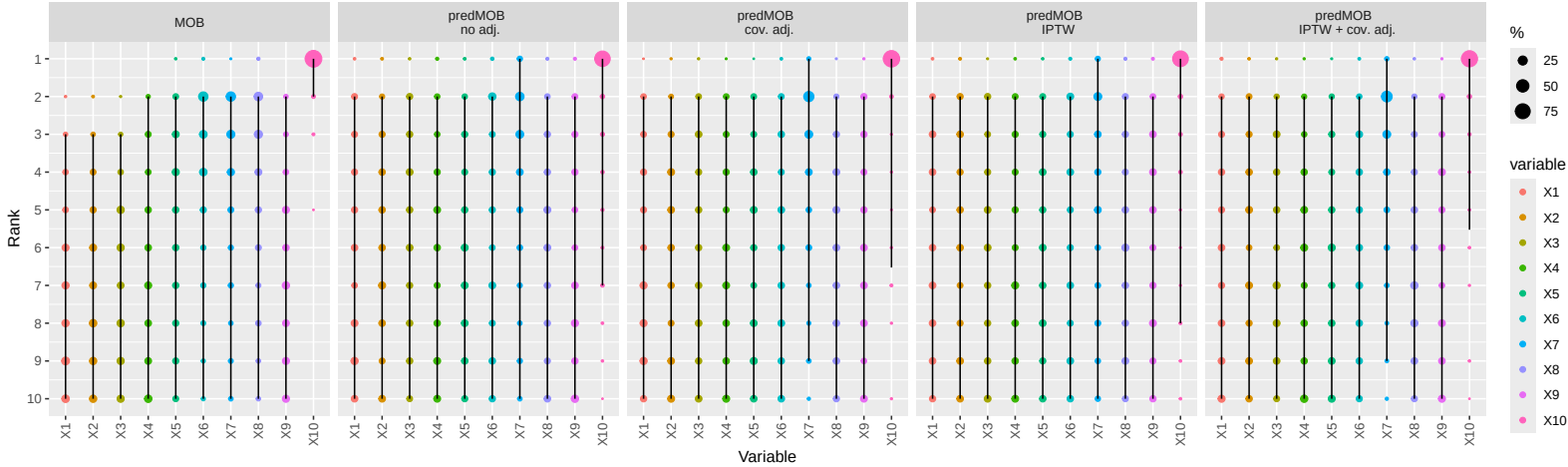
Scenario F

$\mu = 0.125 \cdot \text{trt} + 0.1 \cdot X_4 + 0.15 \cdot X_5 + 0.2 \cdot X_6 + 0.25 \cdot X_7 + 0.2 \cdot X_8 + 0.1 \cdot X_9 + 0.15 \cdot X_{10} + 0.25 \cdot X_{10} \cdot \text{trt}$

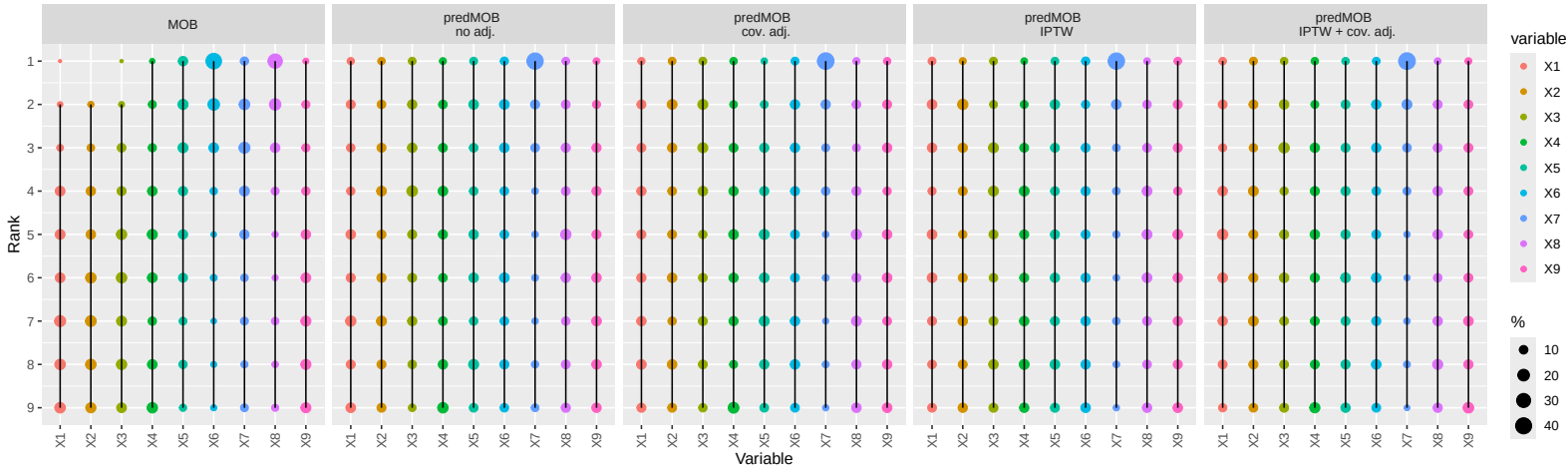


Scenario G1

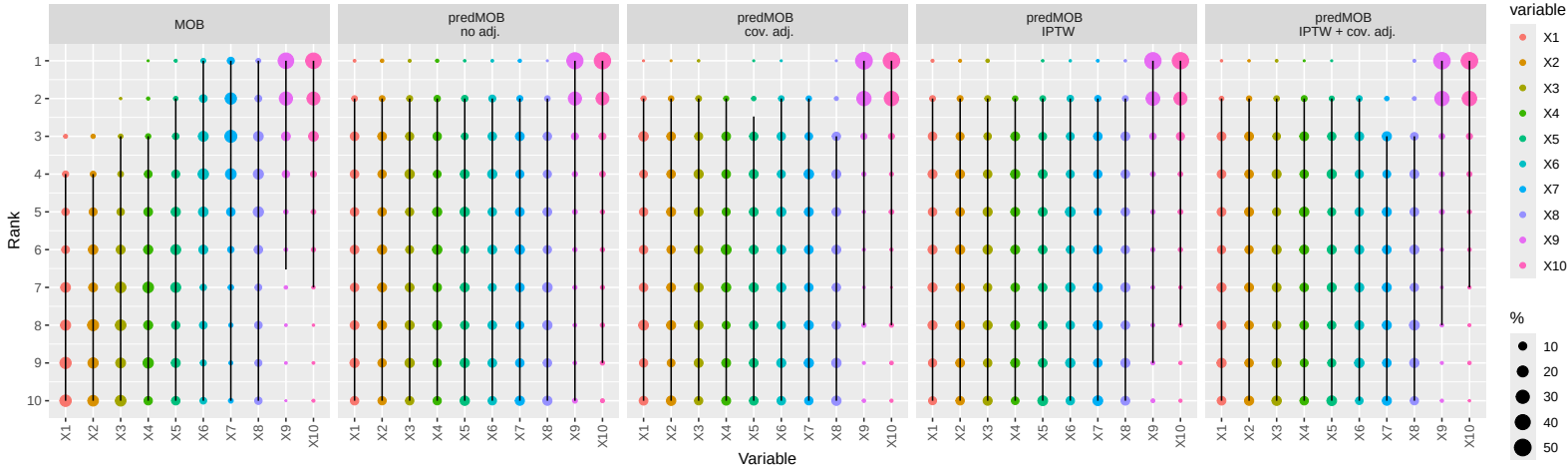
$\mu = 0.25 \cdot \text{trt} + 0.1 \cdot X_4 + 0.15 \cdot X_5 + 0.2 \cdot X_6 + 0.25 \cdot X_7 + 0.2 \cdot X_8 + 0.1 \cdot X_9 + 0.15 \cdot X_{10} + 0.5 \cdot X_{10} \cdot \text{trt}$



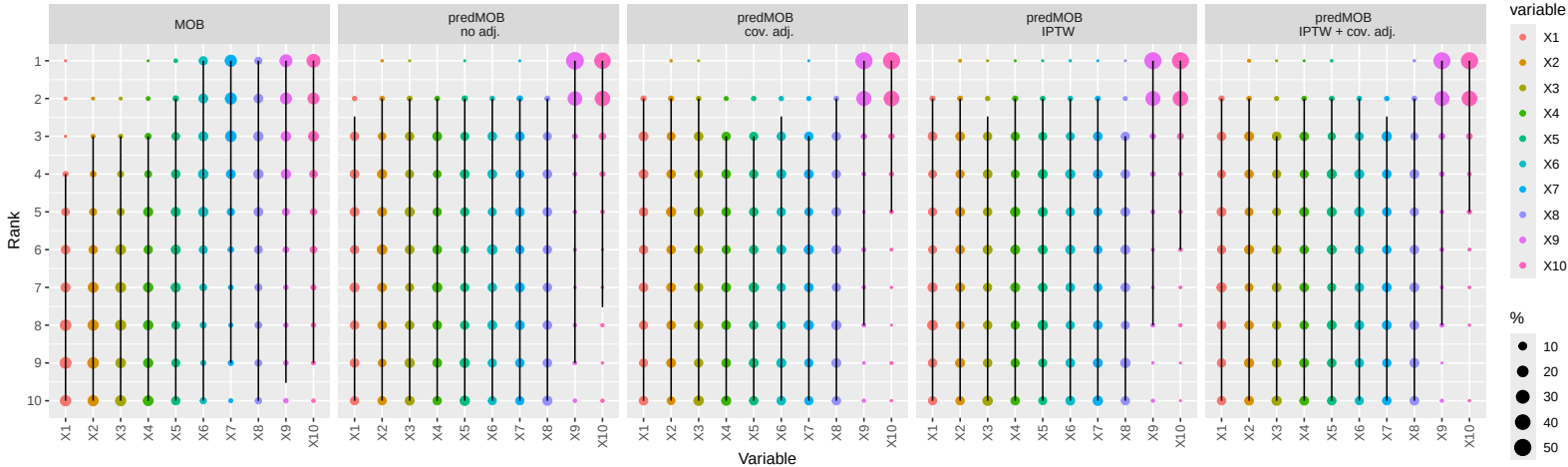
Scenario G2  
 $\mu = 0.25 \cdot \text{trt} + 0.1 \cdot X_4 + 0.15 \cdot X_5 + 0.2 \cdot X_6 + 0.25 \cdot X_7 + 0.2 \cdot X_8 + 0.1 \cdot X_9 + 0.15 \cdot X_{10} + 0.5 \cdot X_{10} \cdot \text{trt}$



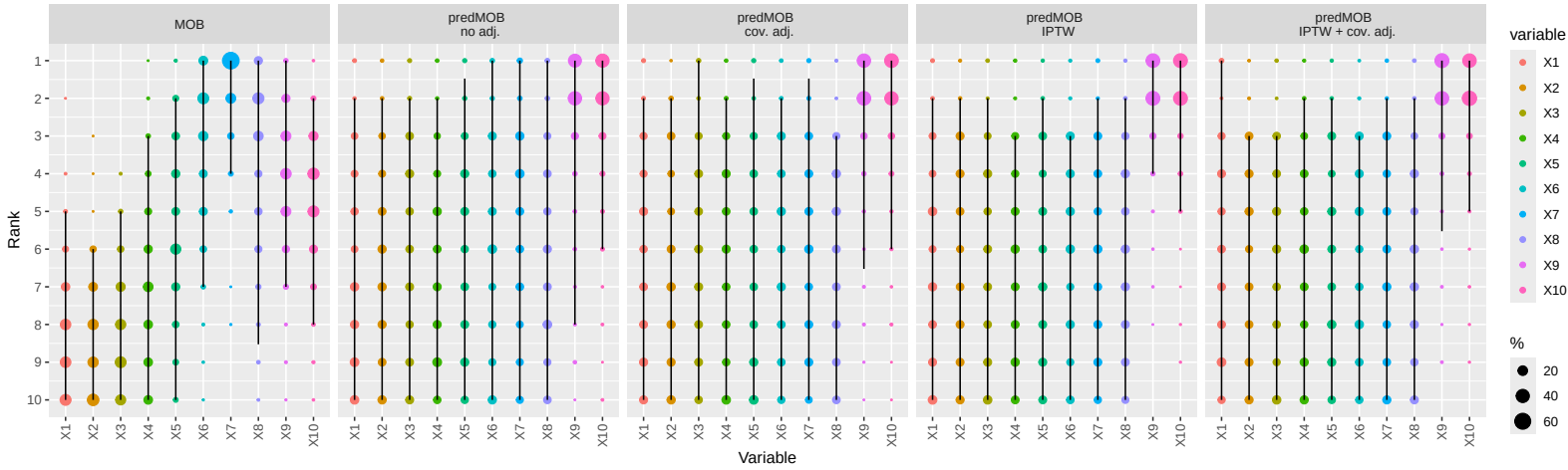
Scenario H1  
 $\mu = 0.25 \cdot \text{trt} + 0.1 \cdot X_4 + 0.15 \cdot X_5 + 0.2 \cdot X_6 + 0.25 \cdot X_7 + 0.2 \cdot X_8 - 0.5 \cdot X_9 \cdot \text{trt} - 0.5 \cdot X_{10} \cdot \text{trt}$



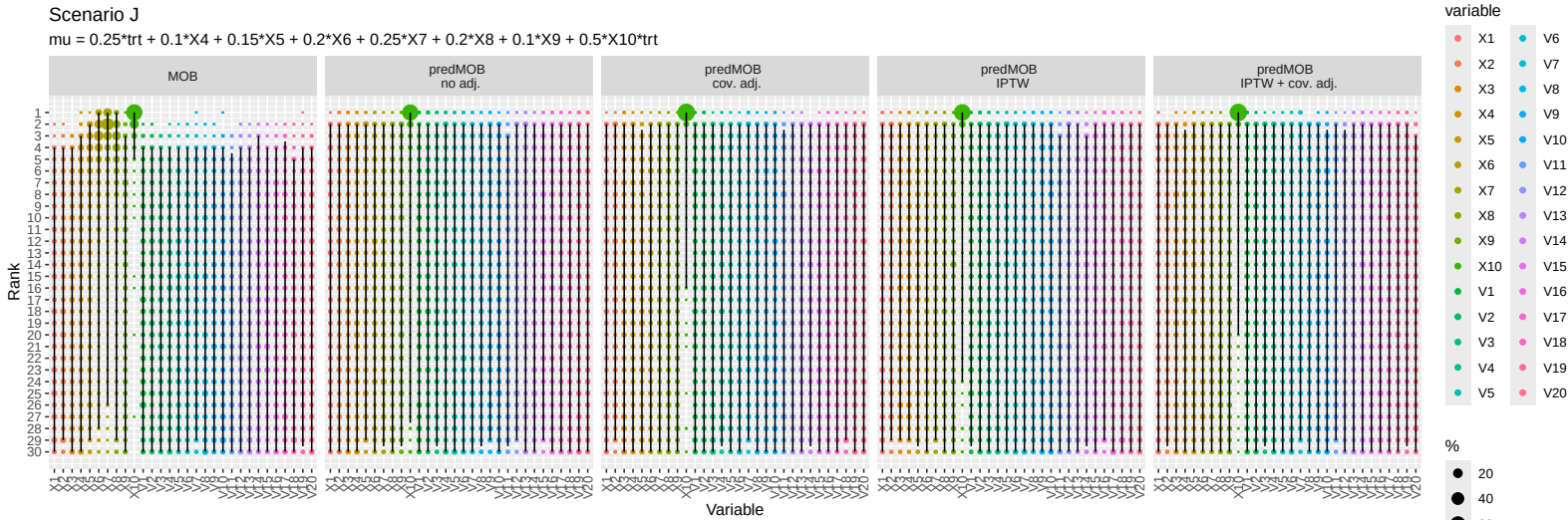
Scenario H2  
 $\mu = 0.25 \cdot \text{trt} + 0.1 \cdot X_4 + 0.15 \cdot X_5 + 0.2 \cdot X_6 + 0.25 \cdot X_7 + 0.2 \cdot X_8 + 0.3 \cdot X_9 + 0.15 \cdot X_{10} - 0.5 \cdot X_9 \cdot \text{trt} - 0.5 \cdot X_{10} \cdot \text{trt}$



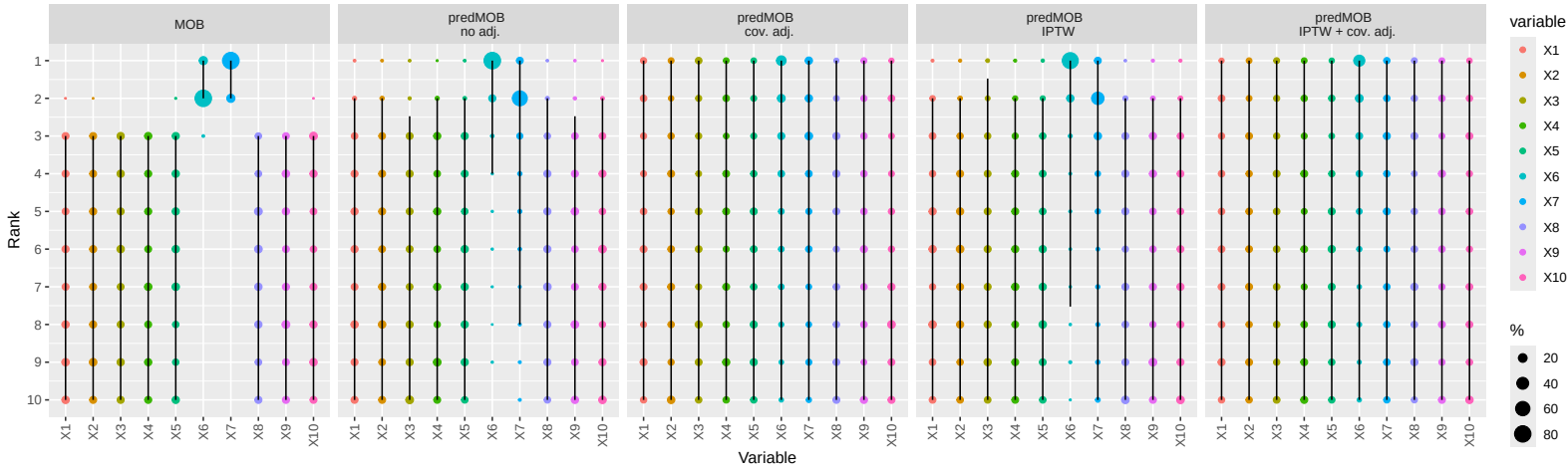
Scenario I  
 $\mu = 0.25 \cdot \text{trt} + 0.1 \cdot X_4 + 0.15 \cdot X_5 + 0.2 \cdot X_6 + 0.25 \cdot X_7 + 0.2 \cdot X_8 + 0.1 \cdot X_9 + 0.5 \cdot X_9 \cdot X_{10} \cdot \text{trt}$



Scenario J  
 $\mu = 0.25 \cdot \text{trt} + 0.1 \cdot X_4 + 0.15 \cdot X_5 + 0.2 \cdot X_6 + 0.25 \cdot X_7 + 0.2 \cdot X_8 + 0.1 \cdot X_9 + 0.5 \cdot X_{10} \cdot \text{trt}$



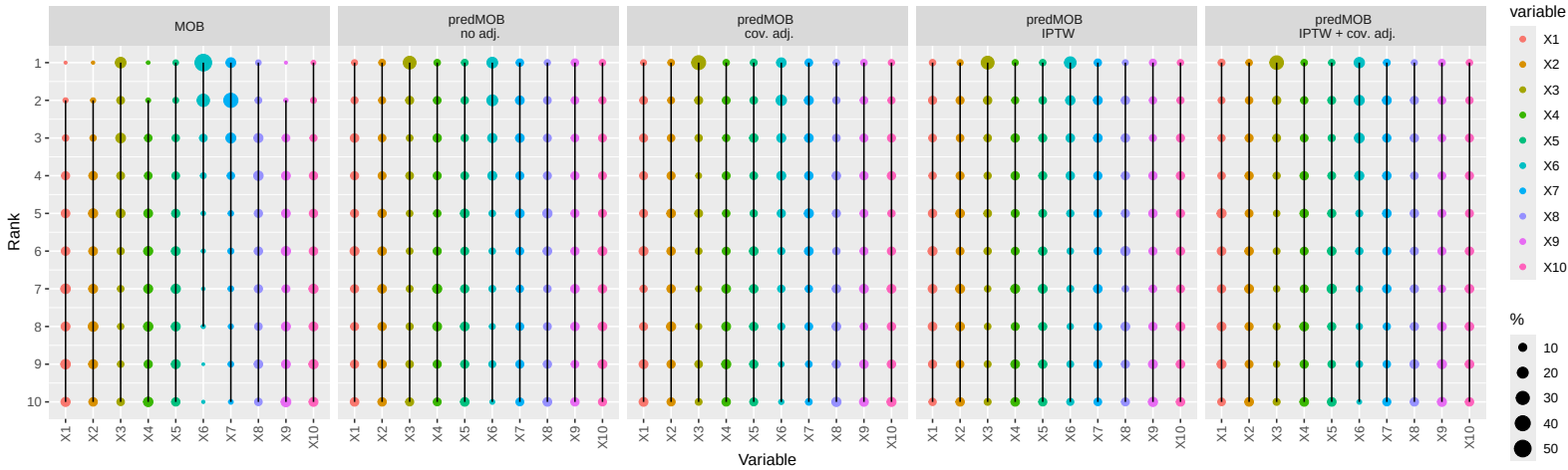
Scenario K  
 $\mu = 0.25 \cdot \text{trt} + 0.8 \cdot X6 + X7$



ddd32

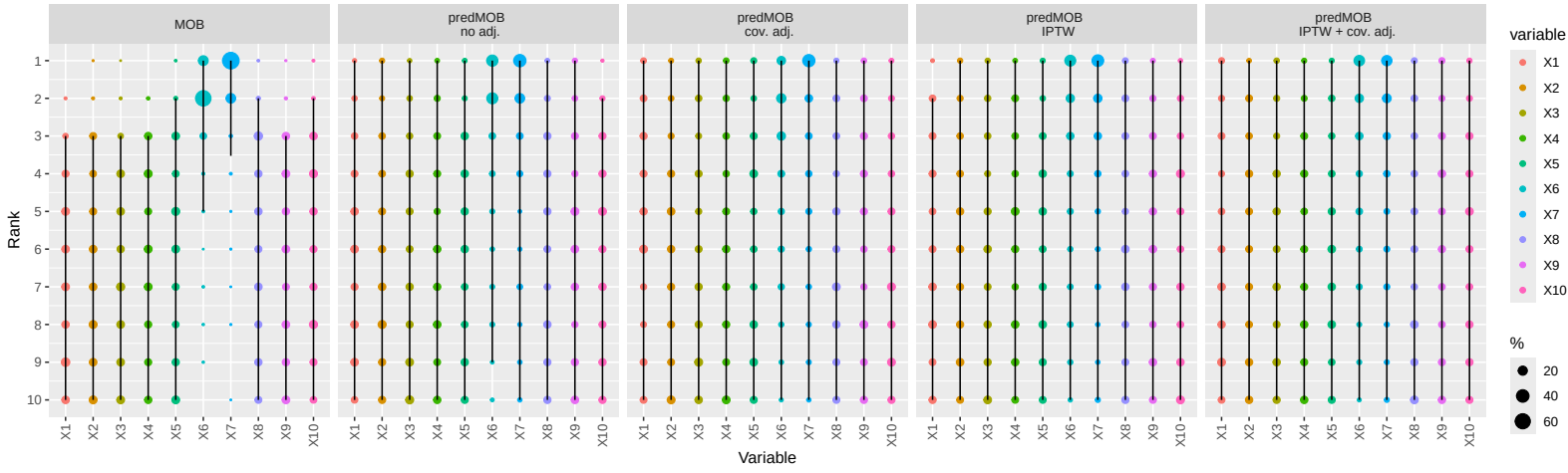


Scenario L  
 $\mu = 0.25 \cdot \text{trt} + 0.1 \cdot X_4 + 0.15 \cdot X_5 + 0.2 \cdot X_6 + 0.25 \cdot X_7 + 0.2 \cdot X_8 + 0.1 \cdot X_9 + 0.15 \cdot X_{10} + 0.5 \cdot X_3 \cdot \text{trt}$

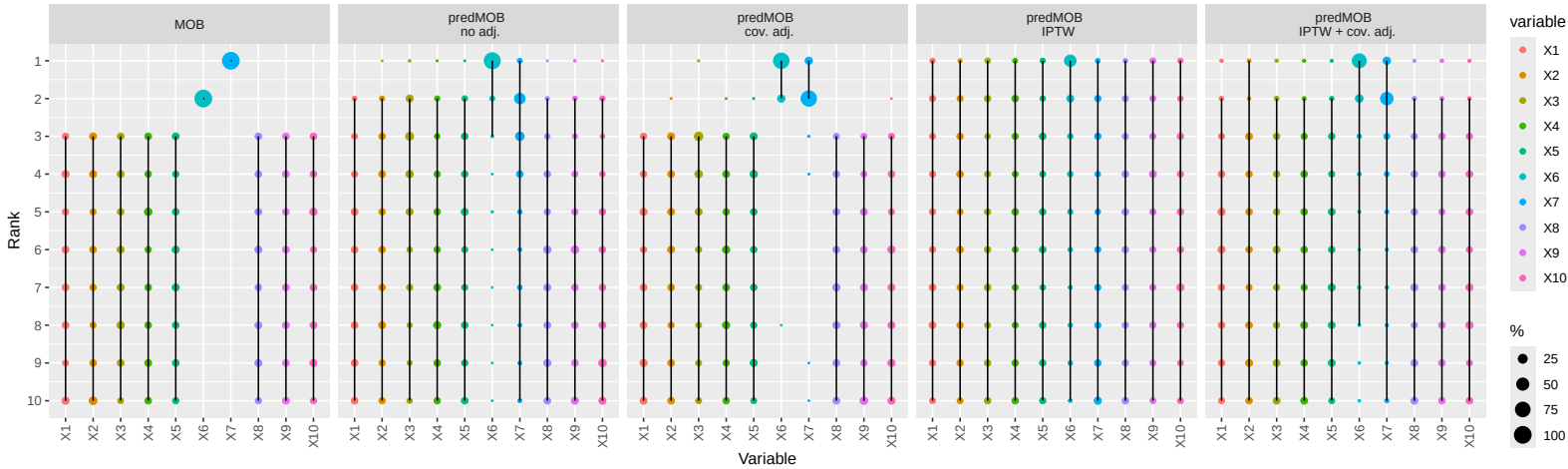


ddd33

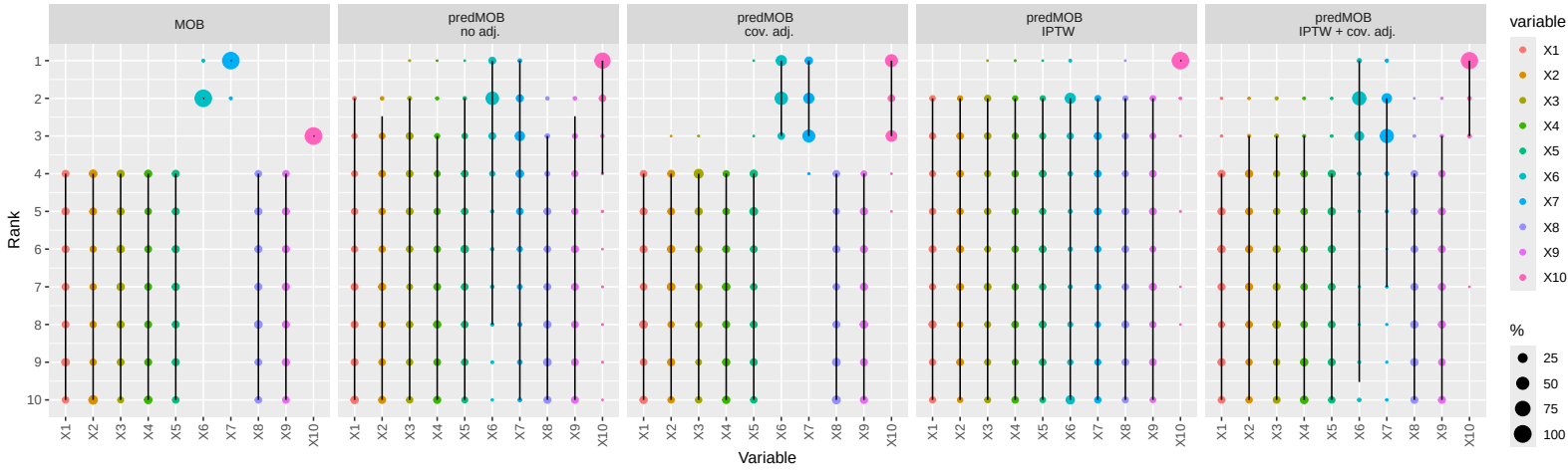
Scenario M  
 $\mu = 0.25 \cdot \text{trt} + 0.1 \cdot X_4 + 0.15 \cdot X_5 + 0.2 \cdot X_6 + 0.25 \cdot X_7 + 0.2 \cdot X_8 + 0.1 \cdot X_9 + 0.15 \cdot X_{10} + 0.5 \cdot X_7 \cdot \text{trt}$



Scenario N  
 $\mu = 0.5 \cdot \text{trt} + 0.75 \cdot \text{abs}(X7 - 1) - 0.8 \cdot \text{abs}(X6 - 1) \cdot X7 + 1.5 \cdot X6 \cdot X7$

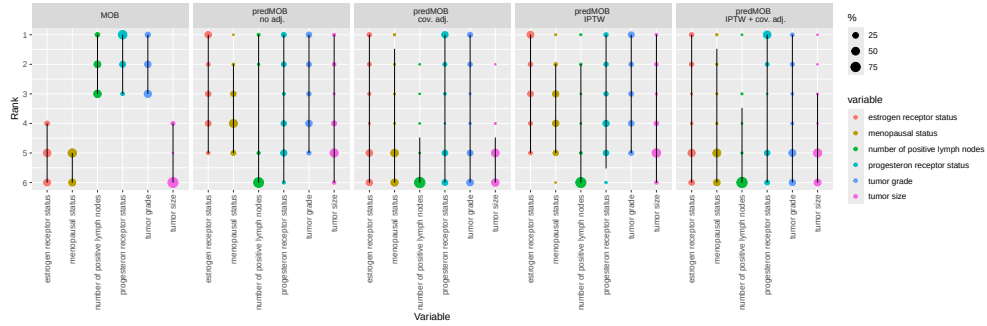


Scenario P  
 $\mu = 0.5 \cdot \text{trt} + 0.75 \cdot \text{abs}(X7-1) - 0.8 \cdot \text{abs}(X6-1) \cdot X7 + 1.5 \cdot X6 \cdot X7 + 1.5 \cdot X10 \cdot \text{trt}$

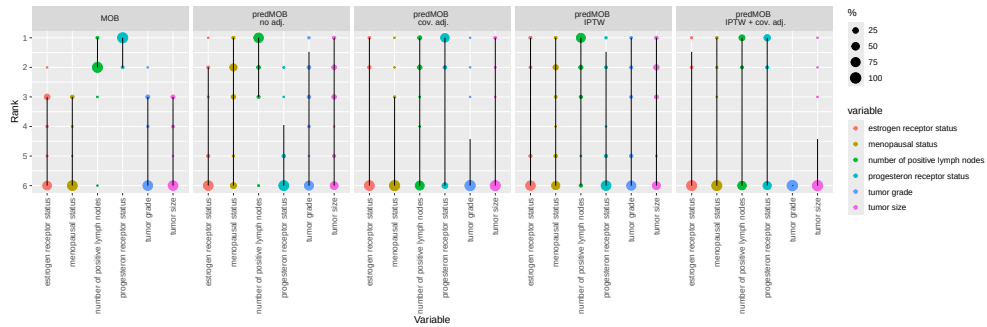


## S 6.2 Application examples

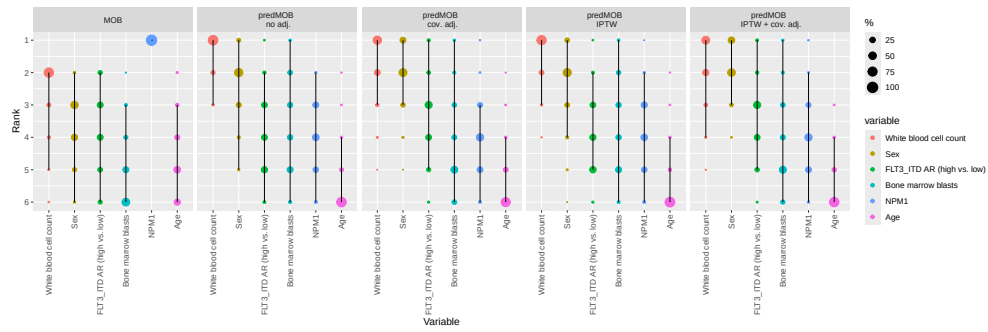
In this section we present the ranking approach to the results of MOB and predMOB for two clinical applications.



**Fig. S.15:** In randomized cohort of the GBSG2 trial the ranking of the permutation importance varies across the predMOB approaches.



**Fig. S.16:** In non-randomized cohort of the GBSG2 trial all predMOB approaches assign high permutation importance rank to number of positive lymph nodes. After adjustment for covariates and IPTW this variable shares top rank along with progesterone receptor status.



**Fig. S.17:** Permutation importance ranking of the different variables using various predMOB approaches show that white blood cell count and sex are consistently ranked first and second, indicating possible predictive effects of these variables in the AMLSG 16-10 trial.